Supplementary Table S4.2 of Chapter 4.

Comorbidities in Juvenile-onset Rheumatic Diseases: a systematic review and meta-analysis

Shall be considered in conjunction with the

Thesis submitted in accordance with the requirements of the

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Supplementary Table S4.2. Studies comparing comorbidity incidence between cases and controls of JIA/jSLE patients reported during childhood and adulthood. NR=p value not reported, NS=P value described as NS 3
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Supplementary Table S4.2. Studies comparing comorbidity incidence between cases and controls of JIA/jSLE patients reported during childhood and adulthood. NR=p value not reported, NS=P value described as NS

Study	Measure of effect	Control group	Comorbidity	Effect size	95% CI	p-value
Childhood/ JIA						
Schenck et al. (2018), Germany[1]	Prevalence ratio	Diabetes registry data	Type 1 diabetes	1.76	(1.34-2.28)	<0.001
Krause et al. (2017), USA[2]	Rate Ratio	Non-JIA comparator subjects. Age-sex matched	Joint surgeries	3.93	(1.18-24.94)	NR
			Non-joint surgeries	1.9	(1.05-3.67)	NR
			Hospitalizations	2.25	(1.04-5.53)	NR
			Hospitalizations for depression	1.41	(0.47-4.70)	NR
Lee et al. (2020), USA[3]	Adjusted Hazard ratios	Healthy children with at least 1 Wellcare visit	Type 1 diabetes	1.81	(1.03–3.17)	NR
			Type 2 diabetes	1.22	(0.79–1.90)	NR
			Composite diabetes	1.4	(0.94–2.09)	NR
	Age, sex, index date-matched Hazard ratios	Healthy children with at least 1 Wellcare visit	Type 1 diabetes	1.89	(1.11-3.19)	NR
			Type 2 diabetes	1.33	(0.88–2.02)	NR
			Composite diabetes	1.52	(1.05–2.21)	NR
	Age, sex, index date-matched Hazard ratios	Children with asthma	Type 1 diabetes	1.46		NR
			Type 2 diabetes	0.95	(0.64–1.43)	NR
			Composite diabetes	1.11	(0.77–1.60)	NR
	Adjusted Hazard ratios	Children with asthma	Type 1 diabetes	1.48	(0.86–2.56)	NR
			Type 2 diabetes	1.01	(0.66–1.54)	NR
			Composite diabetes	1.18	(0.80–1.73)	NR
Beukelman et al. (2012), USA[4]	Incidence rate ratio	Children with ADHD	Coccidioides	101	(8.1–5319)	NR

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			Salmonella	3.8	(1.20–9.50)	NR
			Herpes zoster	2.1	(1.40–3.00)	NR
			All infections	2.4	(1.70-3.30)	NR
			All infections except Herpes zoster	4.2	(2.10-7.60)	NR
Berthold et al. (2022), Sweden[5]	Hazard ratios	Non-JIA comparator subjects	**Females**			
		Sex, year of birth and residential region matched	Depression	1.1	(0.90–1.50)	NS
			Anxiety	1.2	(0.90–1.50)	NS
			Males			
			Depression	0.8	(0.50–1.40)	NS
			Anxiety	0.6	(0.40–1.10)	NS
Lovell et al. (2021), USA[6]	Log risk ratio	General paediatric patients without JIA	Addison's disease	1.65	(1.02-2.21)	P<0.05
			Autoimmune/Hashimoto's thyroiditis	2.25	(1.72 to 2.71)	P<0.05
			Graves' disease	0.38	(–2.77-2.26)	NS
			Morphoea	3.15	(2.54-3.70)	P<0.05
			Multiple sclerosis	1.9	(–1.27-3.83)	NS
			Takayasu's arteritis	1.45	(–1.71-3.35)	NS
			Type 1 diabetes	1.25	(0.75-1.70)	P<0.05
			Vitiligo	1.76	(0.74-2.59)	P<0.05
			Autoimmune disease not elsewhere classified	2.55	(1.78-3.21)	P<0.05
			Chronic glomerulonephritis	0.73	(–2.45-2.64)	NS
			Chronic urticaria	0.58	(0.28-0.84)	P<0.05
			Episcleritis/scleritis	3.18	(2.14-4.03)	P<0.05
			Erythema nodosum	1.12	(–2.03-3.02)	NS
			Haemolytic anaemia	–0.36	(–3.49-1.54)	NS
			Immune thrombocytopenia purpura	0.4	(–1.42-1.68)	NS
			Leukocytoclastic vasculitis	4.08	(2.19-5.49)	P<0.05

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			Pulmonary fibrosis/interstitial lung disease	2.46	(2.28-2.62)	P<0.05
			Raynaud's syndrome	3.37	(2.80-3.86)	P<0.05
			Sjögren's syndrome/sicca syndrome	3.27	(2.40-5.01)	P<0.05
			Thrombocytopenia purpura	0.57	(–2.61-2.46)	NS
			Vasculitis	3.71	(2.90-4.37)	P<0.05
Simon et al. (2020), USA[7]	Odd Ratio (OR)	Patients with ADHD 1:5 matched for age (± 2 years), sex, number of medical encounters (in-patient, out-patient, and emergency room visits), and calendar year of diagnosis (± 1 year)	**MarketScan**			
			Addison disease	6.16	(2.83-13.42)	NR
			Alopecia areata	2.03	(0.93-4.46)	NR
			Celiac disease	3.99	(2.66-5.98)	NR
			Chronic glomerulonephritis	Infinite		NR
			Chronic urticaria	3.4	(2.74-4.22)	NR
			Crohn's disease	16.84	(10.91-25.99)	NR
			Erythema nodosum	12.05	(3.20-45.4)	NR
			Giant cell arteritis	4.51	(0.28-72.18	NR
			Grave's disease	3.23	(1.43-7.27)	NR
			Hashimoto's thyroiditis	3.57	(2.48-5.14)	NR
			Haemolytic anaemia	6.9	(3.98-11.97)	NR
			Immune thrombocytopenia	4.52	(1.88-10.86	NR
			Interstitial lung disease	24.86	(5.51-112.15)	NR
			Morphoea	21.36	(9.45-48.29)	NR
			Multiple sclerosis	2.26	(0.21-24.90)	NR
			Myasthenia gravis	4.51	(0.64-32.06)	NR
			Pernicious anaemia	0		NR
			Polymyalgia rheumatica	Infinite		NR
			Polymyositis	Infinite		NR
			Primary biliary cirrhosis	0		NR
			Psoriasis	8.42	(5.54-12.78)	NR
			Raynaud's syndrome	5.7	(3.44-9.46)	NR
			Scleritis and episcleritis	17.19	(6.42-46.04)	NR

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		Sjogren's syndrome/sicca syndrome	Infinite		NR
		Systemic sclerosis	24.89	(8.57-72.23)	NR
		Type 1 diabetes	1.7	(1.27-2.27)	NR
		Ulcerative colitis	8.34	4.77-14.59)	NR
		Uveitis	165.1	(108.97-250.14)	NR
		Vasculitis	22.64	(9.42-54.42)	NR
		Vitiligo	2.85	(1.39-5.88)	NR
	Patients with ADHD	**Pharmetrics Data**			
		Addison disease	2.51	(1.16-5.43)	NR
		Alopecia areata	2.43	(1.274.65)	NR
		Celiac disease	3.82	(2.61-5.60)	NR
		Chronic glomerulonephritis	31.57	(3.88-256.56)	NR
		Chronic urticaria	2.04	(1.65-2.51)	NR
		Crohn's disease	12.99	(8.78-19.2)	NR
		Erythema nodosum	9.02	(2.72-29.96)	NR
		Giant cell arteritis	Infinite		NR
		Grave's disease	1.69	(0.79-3.64)	NR
		Hashimoto's thyroiditis	4.26	(3.03-6.01)	NR
		Haemolytic anaemia	4.52	(2.50-8.16)	NR
		Immune thrombocytopenia	4.89	(2.23-10.72)	NR
		Interstitial lung disease	12.03	(3.19-45.33)	NR
		Morphoea	18.97	(7.15-50.33)	NR
		Multiple sclerosis	3.38	(0.76-15.11)	NR
		Myasthenia gravis	Infinite		NR
		Pernicious anaemia	0		NR
		Polymyalgia rheumatica	9.02	(0.82-99.43)	NR
		Polymyositis	22.55	(2.63-193.01)	NR
		Primary biliary cirrhosis	Infinite		NR
		Psoriasis	5.89	(4.07-8.53)	NR
		Raynaud's syndrome	12.16	(7.45-19.84	NR
		Scleritis and episcleritis	13.55	(5.76-31.88)	NR

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			Sjogren's syndrome/sicca syndrome	72.24	(9.58-544.36)	NR
			Systemic sclerosis	90.34	(12.12-673.22)	NR
			Type 1 diabetes	1.82	(1.37-2.41)	NR
			Ulcerative colitis	6.49	(3.81-11.05)	NR
			Uveitis	133.23	(92.47-191.95)	NR
			Vasculitis	10.32	(4.24-25.08)	NR
			Vitiligo	2.61	(1.24-5.49)	NR
Horneff et al. (2016), Germany[8]	Risk ratios	German Childhood Cancer Registry report 2015				
			Methotrexate Drug Cohort			
			All malignancy	3	(0.75-12.0)	0.12
			Lymphoma & Leukaemia	6.7	(1.67-26.82)	0.007
			Etanercept Drug Cohort			
			All malignancy	6.3	(3.00-13.20)	<0.0001
			Lymphoma & Leukaemia	8	(2.99-21.38)	<0.0001
Adulthood / JIA						
Raab et al. (2013), Germany[9]	% difference (absolute)	Community sample in Germany	Cardiovascular diseases (as split below)	1.60		0.343
		Age and gender matched by propensity scores	Hypertension	-0.60		0.759
			Cardiac insufficiency	-0.90		0.111
			Asthma	-2.50		0.098
			Gastrointestinal diseases	-6.40		<0.001
			Gastrointestinal ulcer	-1.00		0.065
			Liver and gallbladder diseases	0.10		0.125
			Renal insufficiency	-1.00		0.064
			Diabetes mellitus	-0.70		0.332
			Depression	-4.10		0.004
Aggarwal et al. (2006), India[10]	% difference (absolute)	Healthy male hospital employees	Osteopenia (Lumbar Spine)	-5.50		NS
			Osteopenia (Hip)	28.30		< 0.05

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			Osteopenia (Distal third of forearm)	10.90		< 0.001
			Osteoporosis (lumbar spine)	22.40		NS
			Osteoporosis (hip)	5.90		< 0.05
			Osteoporosis (Distal third of forearm)	35.70		< 0.001
Smith et al. (2019), USA[11]	Unadjusted Risk ratio	Women without autoimmune disease who delivered live born infants	Pre-eclampsia	2.48	(1.30–4.74)	P<0.05
			Pregnancy induced hypertension without pre-eclampsia	1.75	(0.83–3.69)	NR
			Placenta previa and/or abruption	1.46	(0.46–4.68)	NR
			Preterm labour	2.03	(1.27–3.23)	P<0.05
			Gestational diabetes	1.66	(0.87–3.16)	NR
			Fever during pregnancy	1.19	(0.89–1.58)	NR
			Caesarean section	1.55	(1.23–1.94)	P<0.05
	Adjusted risk ratio	Adjusted risk ratio computed using Poisson regression	Pre-eclampsia	1.87	(0.71–4.96)	NR
			Pregnancy induced hypertension without preeclampsia	0.60		(0.08–4.78)
			Gestational diabetes	0.91	(0.25–3.38)	NR
			Fever during pregnancy	2.61	(1.21–5.63)	P<0.05
	Adjusted risk ratio	Multivariable adjustment for the prior factors	Pre-eclampsia	2.19	(0.82–5.88)	NR
			Gestational diabetes	0.86	(0.24–2.97)	NR
			Fever during pregnancy	3.06	(1.42–6.56)	P<0.05
Simon et al. (2020), USA[12]	Odds ratios	MarketScan patients with ADHD 1:5 matched age (± 2 years), sex, number of medical encounters (in-patient, out-patient, and emergency	Addison disease	4.87	(3.09-7.66)	NR
			Alopecia areata	1.09	(0.45-2.64)	NR
			Coeliac disease	1.32	(0.73-2.39)	NR
			Chronic glomerulonephritis	2.36	(0.59-9.43)	NR

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	room visits), and calendar year of diagnosis (± 1 year).	Chronic urticaria	2.53	1.94-3.30)	NR
		Crohn's disease	3.88	(2.93-5.16)	NR
		Erythema nodosum	0.94	(0.21-4.30)	NR
		Giant cell arteritis	Infinite		NR
		Grave's disease	2.15	(1.26-3.65)	NR
		Hashimoto's thyroiditis	1.56	(1.09-2.25)	NR
		Haemolytic anaemia	6.86	(3.88-12.14)	NR
		Immune thrombocytopenia	3.97	(2.35-6.69)	NR
		Interstitial lung disease	11.54	(6.94-19.2)	NR
		Morphoea	3.63	(1.59-8.28)	NR
		Multiple sclerosis	1.50	(0.96-2.33)	NR
		Myasthenia gravis	2.36	(0.59-9.43)	NR
		Pernicious anaemia	1.30	(0.75-2.27)	NR
		Polymyalgia rheumatica	8.20	(4.34-15.5)	NR
		Polymyositis	7.55	(2.47-23.09)	NR
		Primary biliary cirrhosis	2.83	(0.68-11.84)	NR
		Psoriasis	2.31	(1.73-3.08)	NR
		Raynaud's syndrome	3.24	(2.33-4.50)	NR
		Scleritis and episcleritis	5.25	(2.78-9.94)	NR
		Sjogren's syndrome/sicca syndrome	6.82	(4.76-9.77)	NR
		Systemic sclerosis	10.80	(4.44-26.26)	NR
		Type 1 diabetes	2.50	(1.98-3.14)	NR
		Ulcerative colitis	3.69	(2.69-5.06)	NR
		Uveitis	18.46	(13.8-24.69)	NR
		Vasculitis	12.13	(7.03-20.94)	NR
	PharMetrics patients with ADHD	Addison disease	4.67	(2.83-7.73)	NR
		Alopecia areata	1.66	(0.81-3.41)	NR
		Coeliac disease	2.63	(1.68-4.11)	NR
		Chronic glomerulonephritis	24.06	(2.81-205.96)	NR
		Chronic urticaria	2.03	(1.55-2.66)	NR
		Crohn's disease	3.23	(2.36-4.42)	NR
		Erythema nodosum	7.70	(2.52-23.55)	NR

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			Giant cell arteritis	3.21	(0.54-19.19)	NR
			Grave's disease	1.45	(0.83-2.54)	NR
			Hashimoto's thyroiditis	1.99	(1.42-2.80)	NR
			Haemolytic anaemia	5.46	(2.73-10.94)	NR
			Immune thrombocytopenic	4.82	(2.51-9.27)	NR
			Interstitial lung disease	12.89	(6.63-25.04)	NR
			Morphoea	4.81	(1.69-13.73)	NR
			Multiple sclerosis	1.52	(0.89-2.58)	NR
			Myasthenia gravis	3.85	(1.03-14.34)	NR
			Pernicious anaemia	1.24	(0.70-2.20)	NR
			Polymyalgia rheumatica	3.78	(1.72-8.34)	NR
			Polymyositis	4.01	(1.22-13.14)	NR
			Primary biliary cirrhosis	2.40	(0.22-26.52)	NR
			Psoriasis	2.93	(2.22-3.86)	NR
			Raynaud's syndrome	2.53	(1.76-3.65)	NR
			Scleritis and episcleritis	3.66	(2.02-6.66)	NR
			Sjogren's syndrome/sicca syndrome	6.86	(4.70-10.02)	NR
			Systemic sclerosis	8.83	(3.26-23.88)	NR
			Type 1 diabetes	2.15	(1.66-2.78)	NR
			Ulcerative colitis	2.41	(1.63-3.57)	NR
			Uveitis	29.54	(21.51-40.58)	NR
			Vasculitis	7.63	(4.29-13.56)	NR
			Vitiligo	2.16	(1.12-4.15)	NR
Ma et al. (2022), Taiwan[13]	Relative risk	Individuals without JIA	Obstructive sleep apnoea	1.936	(1.26-2.97)	NR
	Adjusted hazard ratios	Adjusted for demographic variables including age, sex, length of hospital stays, and comorbidities at baseline.	Obstructive sleep apnoea	1.922	(1.24–2.97)	NR
	Adjusted hazard ratios	Adjusted for demographic variables including age, sex, length of hospital stays, and comorbidities at baseline and allergic rhinitis	Obstructive sleep apnoea	1.363	(0.70-2.65)	NS

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Zhang-Jian et al. (2020), Taiwan[14]	Adjusted odds ratio (for infant sex, Charlson comorbidity index, urbanization, income, occupation, maternal nationality)	Non-JIA patients 1:5 matched maternal age and birth year To avoid possible confounding factors and for case control selection, only the first child from either group was included for further analysis.	Stillbirth	0.92	(0.32–2.69)	NR
			Foetal distress	0.96	(0.65–1.41)	NR
			Foetal abnormalities, any	0.91	(0.64–1.29)	NR
			Pre-eclampsia	1.42	(0.82–2.46)	NR
			Gestational diabetes	0.75	(0.48–1.16)	NR
			Pregnancy-related hypertension	1.24	(0.79–1.96)	NR
			Antepartum haemorrhage	1.09	(0.81–1.47)	NR
			Severe postpartum haemorrhage	1.20	(0.77–1.87)	NR
			Chorioamnionitis	1.34	(0.49–3.66)	NR
			Caesarean delivery	1.07	(0.89–1.27)	NR
			Low birth weight <2500g	1.35	(1.02–1.79)	<0.05
			Prematurity <37 weeks	1.19	(0.87–1.64)	NR
Krause et al. (2017), USA[2]	Rate ratio	Non-JIA comparator Age and sex matched	Joint surgeries	8.5	(2.27-120.10)	NR
			Non-joint surgeries	1.92	(0.89-4.57)	NR
			Hospitalizations	1.79	(0.69-5.25)	NR
			Hospitalizations for depression	0.26	(0.02-1.11)	NR
			Pregnancies	0.68	(0.38-1.20)	NS
Childhood / jSLE						
Frittoli et al. (2016), Brazil[15]	% difference (absolute)	Patients with cognitive dysfunction. Gender, age and socioeconomic status matched	Cognitive dysfunction	4.86		0.080
Gad et al. (2013), Egypt[16]	% difference (absolute)	Healthy Children. Age and sex matched	Sensorineural hearing loss	25.00		NR
			Conductive hearing loss	0		NR
Karve et al. (2012), USA[17]	% difference (absolute)	Comparator without IBD or SLE. Age (calculated as of year 2005) and sex, and with a minimum health plan enrolment of 6 months before and 12 months after the assigned	Asthma	8.50	NA	NR
			Atopic dermatitis	0.70	NA	NR
			Allergic rhinitis	2.90	NA	NR
			Thyroiditis	0.70	NA	NR

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		index date of the corresponding patient were matched.	Fracture in post-index period	2.10	NA	NR
			Psoriasis	0.00	NA	NR
			Type 1 diabetes	-0.70	NA	NR
			General diabetes	0.00	NA	NR
			Multiple sclerosis	1.40	NA	NR
			Other CNS demyelinating disease	0.00	NA	NR
			Vitiligo	0.00	NA	NR
			Rheumatoid arthritis	19.7	NA	NR
Sinicato et al. (2017), Brazil[18]	% difference (absolute)	Volunteers with no history of chronic, cardiovascular, and autoimmune disease. Age, gender, and demographic background matched	High LDL-c levels	33.00		0.300
			Hypercholesterolemia	16.17		0.210
			Insulin resistance	4.90		0.177
			Arterial hypertension	9.80		0.074
			Albumin excretion	12.20		0.080
Adulthood/ jSLE						
Sinicato et al. (2017), Brazil[18]	% difference (absolute)	Volunteers with no history of chronic, cardiovascular, and autoimmune disease. Age, gender, and demographic background matched	Insulin resistance	0		0.141
			Hypertension	11.4		0.086
			Albumin excretion	11.4		0.086

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